

PROBLEM NO PROBLEM

Reader Guide to the paper, the MATHIC, the evidence and the 3E6B instrument

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Companion README | September 2026 | Locked v1.0

Read this first

This is the reader's way into the PNP package. It explains which files to open, what the construction does, which statements have machine-checked support, and how the working record can be reconstructed. The paper carries the argument. This guide supplies the map and the vocabulary.

The central result is exact existential elimination over finite Boolean states. The separate question is whether those states can be represented and updated at polynomial cost on a standard machine. The package establishes the stated semantic result; the required general complexity bounds remain open.

Packet position

Problem No Problem is the complexity-theory submission in my research program. It can be reviewed on its own. The framework terminology belongs here so the paper does not have to introduce the whole workshop before it turns a screw.

Reading routes

Page	What you will find
2	PNP orientation, the current-state fold and the vocabulary
3	Development timing and the drift record
4	The artifact map and current edition
5	Theorems, finite replay and open obligations
6	Shared framework reference for MATHIC, TRYX and ENIAD
7	Human-directed AI work and the accountability scoreboard
8	Reproduction, sources and review instructions

LOCKED records a fixed artifact identity. Mathematical status is a separate field. A file can stop moving before every question about it has been answered.

“Trust is welcome. Receipts are forever.”

PART I PNP READER ORIENTATION

1 What the title means

Problem No Problem puts the question and the proposed operation side by side. I begin with the Boolean state in hand, separate one coordinate into its two possible values, and combine the results. Each completed fold becomes the state for the next one. The title invites inspection; the theorem states what survives it.

2 One move and its successor

Let a be the restriction of the current state f at $x = 0$, and b its restriction at $x = 1$. For Boolean values, the existential fold is:

$$E_x(f) = a \vee b = a + b - ab$$

The result is true exactly when at least one branch is true. Repeating this operation removes the selected coordinates until one Boolean remains. That terminal value answers whether the original finite Boolean function has a satisfying assignment. Every step acts on the successor just produced.

The paper develops the algebra, four supporting lemmas and the finite semantic closure theorem. Section 4.1 gives the compact formula progression. Section 7 states the additional representation and operation bounds needed for a polynomial-time conclusion.

3 Vocabulary before machinery

Term	Meaning in this package
Current state	The function now in force after the preceding committed operation.
Restrictions	The same current function evaluated with the selected coordinate fixed at 0 or 1.
Mirror closure	The record of shared contribution and unmatched residue before the fold.
Existential fold	Combine the restrictions with Boolean OR and eliminate that coordinate.
Successor	The resulting function used by the next operation.
Local closure	The terminal semantic result within the declared finite setting.
3E6B	The triangular cipher and formula inspection instrument.

4 Choose a route

For the argument, read the paper and its references. For construction order, inspect the locked M0–M9 MATHIC. For machine checking, read the named Lean statements and their scope. For executable evidence, run the replay and independent checker. The instrument is useful when you want to watch a displayed operation and inspect its formula.

DEVELOPMENT TIMING AND DRIFT

5 Keep the clocks separate

I keep build records because memory is a poor version-control system. Contracts, source identities, receipts and commits let a later reader reconstruct the sequence. They also expose the time spent getting back to an instruction that should have survived the first time.

Work phase	Recorded measure	What the number means
Original PNP development	≈53h 22m 43s	Reconstructed elapsed time from the active build to original internal/local closure.
E6B through repaired 3E6B v65	71h 24m 40s	Earliest recovered saved E6B build to the v65 repair commit; includes idle time and redesign.
3E6B regression recovery	1h 56m 11s	One audit-to-repair-commit interval covering six defect areas.
Later PNP paper production	3h 30m	My reported lost-time block; independently timed start and end are unavailable.

The original PNP clock runs from approximately August 4, 2026, 16:11:17 AKDT to August 6, approximately 21:34 AKDT. The endpoint comes from the original execution package. Later recovery, Lean work and paper production do not get added to that original research interval.

The instrument window starts with saved E6B v0.1 on September 14 at 07:31:06.664454 UTC and ends at the September 17 v65 repair commit at 06:55:47 UTC. It includes the circular instrument's triangular successors. It is an observed delivery window, not a recovered first-keystroke time or a claim of 71 hours of hands-on labor.

6 What the drift record captures

AI drift is the worst part of the process, but it is manageable. The September 17 instrument audit identified plain-text rejection, Keyless Decode direction, topology placement and ownership, star reset, Paste Crypto visibility, and stale receipt parameters. The repair committed at 06:55:47 followed the audit response at 04:59:36. Six defects share that one interval; multiplying the hours by six would be creative accounting.

My 3½-hour report concerns later PNP paper recovery. It does not measure drift during the original August research. Requested features and intentional redesigns are not automatically failures. Total active labor and whole-project drift remain unmeasured where the records lack session boundaries. Unknown is not zero.

[Read the original timing receipt and current work-metrics record](#)

THE ARTIFACT MAP

7 Which file answers which question

Artifact	Its job	Where to begin
Locked PNP MATHIC	Canonical order, measure names and scored construction.	M0–M9 score under charts/npn.
Paginated paper	Conventional argument, formula progression, four figures and references.	18 physical pages: splash plus numbered pages 1–17.
Governing source	Preserved source for the construction and its provenance.	Continuity reintroduction benchmark under publications/npn-mathic/sources.
Lean project	Formal statements and proof-checking instructions.	PNP.md and TryxProof.lean under verification/tryx-lean.
Replay and independent checker	Finite execution, intermediate states and negative control.	check_package.py and the two result records.
3E6B v65	Interactive cipher and formula inspection.	Successor to the archived v64 instrument.
Figures and splash	Explanatory presentation and visual identity.	Included in the locked paper; separate figure identities remain preserved.
README and receipts	Reading routes, vocabulary, chronology and artifact identity.	This guide and the repository evidence records.

8 Current paper edition

The audited edition places Figure 1 on page 2, Figure 2 on page 4, the formula progression in §4.1 on page 9, Figure 3 on page 12, Figure 4 on page 15, and references on page 17. The splash is unnumbered. This map uses the paper’s printed page numbers, not this guide’s pages.

[Open the locked paginated paper and its editable-source archive](#)

The edition archive is in the project repository and requires its access permission. The standalone public repository supplies the source, formal checks and replay package. Its earlier Markdown manuscript is not an exact copy of the latest typeset edition.

9 Identity before replacement

Use the edition checksum manifest when restoring the paper. The editable DOCX is a layout source; the stored rendering inputs and finalization script also supply the final splash, page numbering, contents links and bookmarks. A fresh edit becomes a successor edition rather than silently replacing locked bytes.

CLAIMS AND EXECUTABLE EVIDENCE

10 What has been checked

Statement or test	Evidence	Scope
Boolean OR equals $a + b - ab$	boolean_fold_algebra	Boolean inputs and their integer encoding.
One hinge preserves existential truth	hinge_exact	The two restrictions of the declared Boolean function.
Repeated elimination gives the terminal answer	resolve_correct	Every finite assignment dimension in the formal semantic model.
Finite implementation agrees with an independent oracle	Replay and independent-check records	Declared three-variable population, initial and intermediate states.
Replacing OR with AND is rejected	Negative control	The checker detects this deliberate corruption.
A compact exact polynomial-time implementation exists	Open obligation	Not supplied by those semantic theorems or finite runs.

11 The finite fixture

Three variables give 26 admissible nonempty, non-tautological clauses. Taking sets of zero through four clauses yields 17,902 formulas. Three coordinate folds per formula produce 53,706 folds. The recorded terminals are 16,241 SAT and 1,661 UNSAT, with zero disagreements in the primary replay.

The independent checker builds clause masks separately and compares direct quantification with the fold engine at the initial, intermediate and terminal states. Agreement with the same evaluator twice would leave an obvious blind spot; the independent construction is there to test it.

“If the checker never says no, you may have earned a golf clap.”

12 What remains to be supplied

The truth-table implementation begins with 2^n entries. A polynomial-time implementation needs a compact exact representation and polynomial bounds for its encoding, updates, normalization and readout in the original input length. Witness reconstruction needs its own analysis if claimed. The paper’s conditional proposition states that implication explicitly.

Lean acceptance applies to the named formal statements. It does not certify every label, interface or cost claim in the surrounding project. The finite theorem and the engineering question are both worth reading carefully.

PART II SHARED FRAMEWORK REFERENCE

13 MATHIC as an execution map

MATHIC is the scored review object I use to expose the mathematical sequence. A reviewer should be able to identify the admitted state, the comparison, the operation, the residue, the successor and the terminal obligation. The paper gives that sequence conventional exposition; it does not silently rename the source to make a paragraph read more smoothly.

Score element	What the reader inspects
Measure	The declared construction stage and its place in the order.
Typed role	Which state, comparison or committed operation a quantity belongs to.
Transition	How an operation produces the next state.
Explanation	Plain-language meaning attached to the actual operation.
Receipt	Which input, output, version and check support a recorded result.
Open gate	The exact obligation that has not yet been discharged.

14 TRYX and the current state

TRYX means Trinary Runtime Index. Value is the current object. Perception is the declared comparison or relation. Action is the committed result, which becomes the next Value. It is the working index I use while constructing and reviewing the mathematics.

$$V_0 \rightarrow P_0 \rightarrow A_0 \rightarrow V_1$$

$$V_n \rightarrow P_n \rightarrow A_n \rightarrow V_{n+1}$$

The point is continuity: preserve which object is being handled, what changed, and what the next step inherits. A result from a different branch cannot quietly replace the state under review because it happens to have a familiar name.

15 ENIAD and the addressed field

ENIAD is the larger address-and-provenance framework described in the project as nine-dimensional mathematical plus zero topology. The recursive three-phase mechanism supplies the local cycle; the broader organization keeps roles, relations and scales distinct. This is framework terminology, not a claim about extra physical dimensions or a shortcut around a complexity bound.

Atomic, Micro and Macro identify scale roles. Squadron describes cross-scale audit or simultaneous presentation. These terms help organize the work; they do not add a fourth proof engine. The PNP semantic theorem can be read from its Boolean definitions without adopting the framework vocabulary.

HUMAN DIRECTED AI AND THE SCOREBOARD

16 John Henry at the railhead

There is a deliberate human-machine challenge underneath this program. My version of John Henry is not a plan to beat artificial intelligence without artificial intelligence. That would be a peculiar way to use the century. I want the machine's output to accumulate into a coherent investigation instead of becoming another pile of plausible fragments.

The useful question is whether the same mathematical object survives the work. Which assumptions entered? Which branch produced the result? What remains unresolved? Can another person reconstruct the transition? Those questions matter more than how quickly the next paragraph arrives.

So yes, there is a gauntlet on the floor. I just prefer that the locomotive help swing the hammer.

17 A challenge needs measurable terms

Measure	Evidence to keep
Continuity	Accepted baseline and the version actually used for each successor.
Source fidelity	Whether formulas, definitions and figure identities remain bound to their source.
Drift frequency	Distinct departures from prior requirements, grouped by recovery episode.
Recovery burden	Detection, repair and acceptance times; active work separated from waiting.
Independent checking	Which tests use a separate construction and which share assumptions.
Unresolved obligations	Open items carried forward without being omitted or promoted.

The current records support a retrospective baseline, not a controlled comparison of human and AI productivity. The two projects do not yet have complete active-time logs. A precise-looking percentage built from incompatible clocks would only make the spreadsheet more confident than the evidence.

18 How I keep the work cumulative

I hold the accepted source fixed, compare the successor against it, and rerun the checks relevant to the change. I keep failed evidence and recovery records so a later pass can explain what happened. A lock gives the next session somewhere definite to stand.

Publishing the applications with their evidence also makes the framework answer to concrete work. A method is useful when another person can inspect what it preserved, not merely when its author can describe what it intended.

“Trust is welcome. Receipts are forever.”

REPRODUCTION AND REVIEW

19 Start with the packet you are reviewing

Use the standalone public PNP repository for reproducibility. Record its commit before running the checks. From the repository root, run:

```
python3 publications/pnp-mathic/check_package.py
```

The package checker verifies tracked hashes and reproduces the replay and independently generated oracle results, including the negative control. The formal verification project has its own pinned toolchain and instructions. Follow that README rather than substituting whichever Lean version happens to be installed.

[Public PNP repository and current README](#)

[Lean scope and verification instructions](#)

[Original PNP development-time receipt](#)

[PNP and 3E6B work-metrics receipt](#)

[3E6B v65 repair and instrument source](#)

20 Status words with specific jobs

Status	Meaning
CLOSED	Established within the explicitly declared scope.
CONDITIONAL	An implication holds under its named assumptions.
LEAN VERIFIED	The specified formal theorem checked in the recorded project.
REPLAY VERIFIED	The declared finite fixture reproduced.
DIAGNOSTIC	An inspection tool or interpretation with no independent theorem weight.
OPEN	An obligation remains unresolved.
LOCKED	A fixed artifact identity for provenance and restoration.

21 A useful challenge

Name the exact statement or operation, identify the artifact and commit, and state what you observed. For a computation, include the command, input and first disagreement. For a proof, identify the definition or inference at issue. For presentation, show the mismatched figure or formula. That gives the next reviewer an object to examine.

This guide explains the package; it introduces no new theorem. The declared finite semantic closure and recorded execution evidence remain the established scope. Compact exact polynomial-time implementation remains the additional obligation. The record should make both easy to find.

END OF READER GUIDE